dropped by the United States, incinerating the Japanese city of Hiroshima on August 6, 1945. With ninety-two protons packed in its nucleus, uranium is widely described as the "largest" naturally occurring element, although trace amounts of larger elements can be found naturally where uranium ore is mined.

If Uranus deserved an element named in its honor, then so did Neptune. Unlike uranium, however, which was discovered shortly after the planet, neptunium was discovered in 1940 in the Berkeley cyclotron, a full ninety-seven years after the German astronomer John Galle found Neptune in a spot in the sky predicted by the French mathematician Joseph Le Verrier after studying Uranus's odd orbital behavior. Just as Neptune comes right after Uranus in the solar system, so too does neptunium come right after uranium in the Periodic Table of elements.

The Berkeley cyclotron discovered (or manufactured?) many elements not found in nature, including plutonium, which directly follows neptunium in the table and was named for Pluto, which Clyde Tombaugh discovered at Arizona's Lowell Observatory in 1930. Just as with the discovery of Ceres 129 years earlier, excitement prevailed. Pluto was the first planet discovered by an American and, in the absence of better data, was widely regarded as an object of commensurate size and mass to Earth, if not Uranus or Neptune. As our attempts to measure Pluto's size became more and more refined, Pluto kept getting smaller and smaller. Our knowledge of Pluto's dimensions did not stabilize until the late 1980s. We now know that cold, icy Pluto is by far the smallest of the nine, with the diminutive distinction of being littler than the solar system's six largest moons. And like the asteroids, hundreds more objects were later discovered in the outer solar system with orbits similar to that of Pluto, signaling the end of Pluto's tenure as a planet, and the revelation of a heretofore undocumented reservoir of small icy bodies called the Kuiper belt of comets, to which Pluto belongs. In this regard, one could argue that Ceres, Pallas, and Pluto slipped into the Periodic Table under false pretenses.

Unstable weapons-grade plutonium was the active ingredient in the atomic bomb that the United States exploded over the Japanese city of Nagasaki, just three days after Hiroshima, bringing a swift end to World War II. Small quantities of non-weapons-grade radioactive plutonium can be used to power radioisotope thermoelectric generators (sensibly abbreviated as RTGs) for spacecraft that travel to the outer solar system, where the intensity of sunlight has diminished below the level usable by solar panels. One pound of plutonium will generate ten million kilowatt-